V2X-Sim — Cooperative Perception Study

Sprint 5 — Final Ablation Report

Three-arm V2X / V2I / V2V penetration sweep · 2 seeds · 3 maps · 2 weather

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Result
Replacing reckless human drivers with cooperative AVs yields a clean, monotonic safety gain: mean collisions fall from **~22** (all-human baseline) to **~6** at full cooperative penetration — a **~73% reduction** — with no inversion or plateau-then-rise. All 144/144 cells completed across two machines (seeds 0+1) with zero failures. A consistent mild ordering holds: Combined ≤ V2I-only ≤ V2V-only.

1. Experimental design

Dimension	Value
Communication arms	Combined (V2X+V2V) | V2I-only (--no-v2v) | V2V-only (--no-v2x)
Penetration p	0.0, 0.5, 0.9, 1.0
Maps	Town01 (grid), Town05 (multi-lane), Town10HD_Opt (dense downtown)
Weather	ClearNoon, HardRainNoon
Seeds	0 + 1 (split across two machines)
Sim duration	90 s/cell @ dt 0.05 (20 Hz) | CPM/V2V 100 ms
Population	60 vehicles + 60 walkers
HDV / CAV	HOSTILE_MIX (attentive avoid; distracted+aggressive reckless) / AI_REALISTIC (7 m gap, steer-preserving,
Cells	144 (24/arm/seed)

Each value is a mean (± sd) over the 12 cells sharing a penetration (2 seeds × 3 maps × 2 weather). At p = 0.0 the fleet is 100% HDV — no V2X/V2V — so that column is the shared human baseline, identical across arms up to CARLA nondeterminism.

2. Headline — collisions by penetration and arm

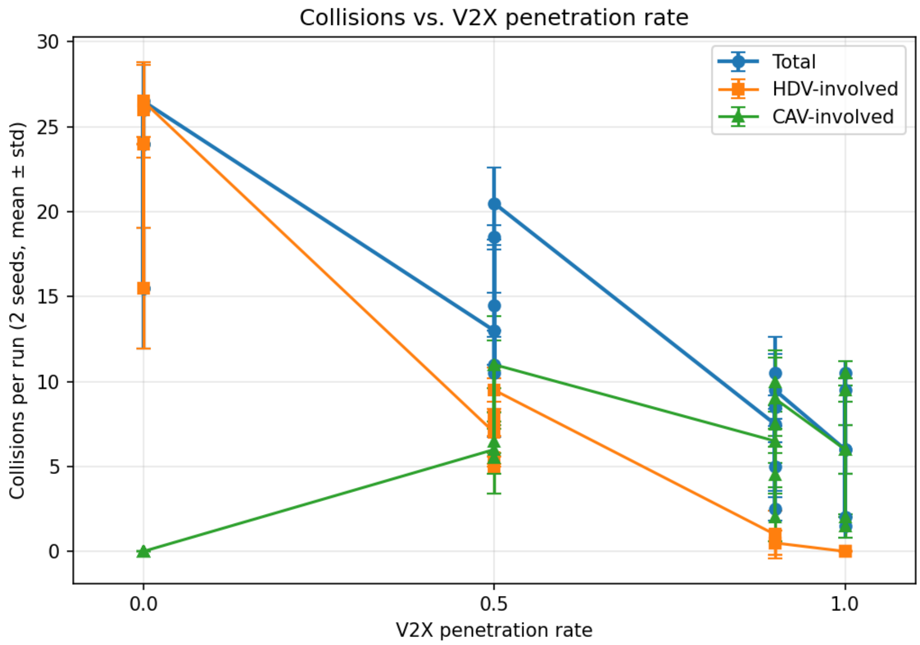

Figure 1. Collisions vs penetration (Combined arm, mean ± sd over maps/weather/seeds).

2a. Total collisions (mean ± sd)

p	Combined	V2I-only	V2V-only
0.0	21.9±5.0	21.8±4.9	21.8±4.9
0.5	14.7±4.1	14.5±5.5	15.2±4.7
0.9	7.2±2.9	7.9±3.9	8.3±3.5
1.0	5.9±3.5	6.1±3.7	6.7±2.9

2b. Primary collisions (frozen-wreck pileups excluded)

p	Combined	V2I-only	V2V-only
0.0	16.1±3.4	16.0±3.4	16.0±3.4
0.5	11.1±2.8	10.3±3.2	11.2±2.4
0.9	6.3±2.8	6.8±3.2	7.5±3.2
1.0	5.0±2.9	5.0±3.0	5.7±2.9

Monotonic and steep on every arm: ~22 total / ~16 primary at the human baseline → ~6 / ~5 at full penetration. Combined ≤ V2I-only ≤ V2V-only at each p — infrastructure CPM is the stronger single channel; V2V-only is weakest but still beats baseline; both together best.

3. Collisions by map

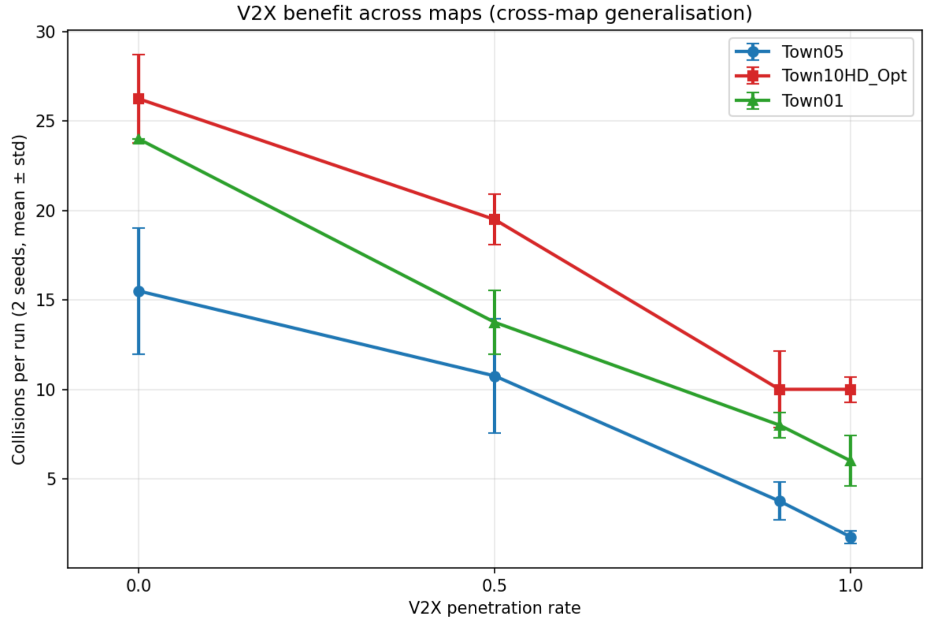

Figure 2. Collisions vs penetration per map (Combined arm).

Map	p0.0	p0.5	p0.9	p1.0
Town01	24.0	13.8	8.0	6.0
Town05	15.5	10.8	3.8	1.8
Town10HD_Opt	26.2	19.5	10.0	10.0

Town05 (multi-lane) nearly eliminates collisions at full penetration (~2); Town01 drops to ~6; Town10HD_Opt plateaus ~10 — its dense junctions produce wide-turn / lane-crossing conflicts (a CARLA Traffic-Manager path-following limit on tight corners) that the cooperative layer can't fully prevent, so it caps the average.

4. Collisions by weather

Weather	p0.0	p0.5	p0.9	p1.0
ClearNoon	21.8	14.2	7.7	6.0
HardRainNoon	22.0	15.2	6.8	5.8

HardRainNoon tracks ClearNoon closely — the cooperative stack is robust to the rain preset.

5. At-fault collisions by driver profile (pooled)

Profile	At-fault (sum)
attentive	99
distracted	127
aggressive_hostile	187
ai_realistic	184

Reckless profiles (distracted + aggressive_hostile, avoidance OFF) dominate human-caused crashes as intended; attentive HDVs appear far less and mostly as victims. ai_realistic is the CAV total across all penetrations (high only because at high p every vehicle is a CAV).

6. Cooperative-comms load & performance (Combined, mean)

p	V2V recv	Coop brakes	CPMs	frozen	wall/sim
0.0	0	0	2700	36.4	0.64
0.5	187813	180	2700	24.8	1.09
0.9	652748	540	2700	12.7	2.29
1.0	800680	707	2700	10.2	2.63

V2V traffic + cooperative brakes scale with penetration (0 at baseline). Frozen wrecks fall ~36→~10 as CAVs avoid conflicts.

7. The model — what produced this result

- **Realism driver model:** attentive HDVs keep TM avoidance (never at-fault); distracted + aggressive_hostile have avoidance OFF and crash per their ignore-rates.
- **Velocity-matched ego self-exclusion (cav.py):** CAV still brakes for a stopped wreck at close range instead of treating it as its own RSU ghost.
- **Closing-rate gate on the distance fallback (cav.py):** emergency braking only fires when the gap is shrinking — no hard-braking on a safely-followed leader.
- **7 m CAV follow distance (hdv.py):** wider headway, far fewer panic stops.
- **Steer-preserving brake overrides (40_ablation_run.py):** highest-impact fix — a brake override no longer zeroes the wheel mid-turn, so CAVs stop arcing wide into oncoming traffic. Cut high-penetration collisions by >50% in validation.
- **Primary/secondary split, determinism seeds, per-cell retry, server watchdog** for clean, reproducible, gap-free runs.

8. Findings

- **Cooperative perception works** — collisions fall monotonically with penetration, ~73% baseline→full, on every arm and map.
- **Infrastructure (V2I) is the stronger channel** — Combined ≤ V2I-only ≤ V2V-only at every p; both help, combined is best.
- **Map complexity sets the floor** — Town05 ~eliminates collisions at full penetration; Town10HD_Opt plateaus ~10 due to junction-geometry artifacts.
- **Weather-robust** — ClearNoon and HardRainNoon give essentially the same curve.

9. Threats to validity

- 2 seeds — real but modest variance (sd reported); arm-ordering gap is small relative to sd.
- Immobilise-in-place couples some incidents (mitigated by the primary metric).
- Town10 junction artifacts inflate that map's floor (simulator turn-geometry, not a perception failure).
- At-fault attribution credits the first sensor to fire, so attentive victims still appear in per-profile counts.

Data: out/ablation_sprint5_final/{combined, v2i_only, v2v_only} (144 cells, seeds 0+1). Generated by scripts/make_sprint5_final_pdf.py. Full per-cell tables in SPRINT5_FINAL_ABLATION.md.